Meeting 18. Random Projections and Streaming

A vector with hundreds of coordinates may be expensive to store, compare, or transmit. Reducing its dimension helps only if the smaller representation preserves what the calculation needs. Here the question is precise: can a much shorter vector preserve distances among a fixed collection of points?

A random linear map gives a useful answer. It does not retain every possible property of the original vectors, and an individual draw can be poor. With enough output coordinates, however, an appropriately chosen distribution gives a high-probability guarantee for all pairs in a fixed finite dataset. The same linear operation also makes a running summary easy to update and combine.

Companion. The Random projections slide deck contains interactive experiments on word vectors, synthetic clouds, and event streams. These notes use the same 200 word vectors, with separately specified NumPy random maps. Errors refer to **squared Euclidean distances** throughout.

The main points

1. **Use one map for all points**

   A shared linear map gives every observation the same new coordinates and maps a difference to the difference of the mapped points. Sections 1-2.

2. **State the error criterion**

   Typical pair error, worst pair error, and neighbor overlap are different measurements. A squared-distance tolerance is not the same numerical tolerance on distance. Section 3.

3. **More dimensions increase reliability**

   The Johnson-Lindenstrauss guarantee controls every pair of a fixed finite set with high probability. It does not preserve every vector in the original space. Section 4.

4. **PCA and random projection serve different objectives**

   High retained variance does not guarantee small relative error for every pair. Small distance errors can also change nearly tied rankings. Section 5.

5. **Linearity enables updates and merging**

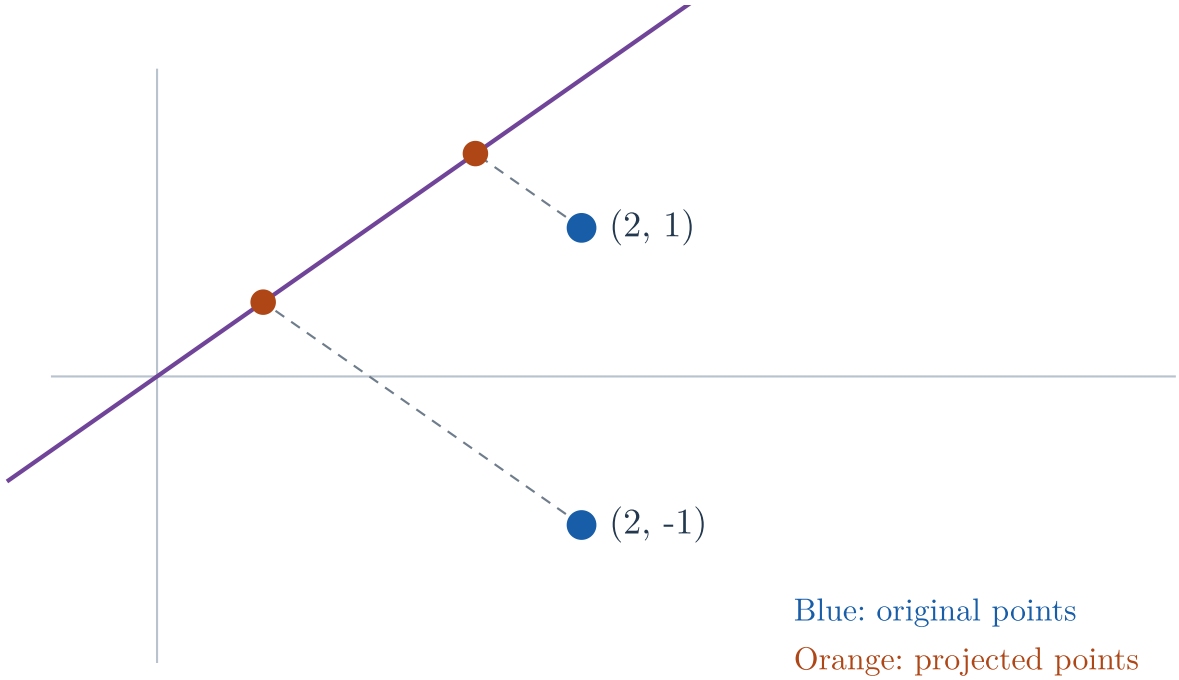
   A sketch can be maintained one event at a time and combined across workers using the same map and coordinate definitions. These algebraic identities are exact; geometric estimates remain approximate. Sections 6-7.

1. Decide what the smaller representation should preserve

One million observations with 300 coordinates require 300 million stored numbers. At float32 precision, four bytes per number, the data values occupy 1.20 GB in decimal units. Keeping 30 coordinates per observation reduces this to 0.12 GB. The map and the computation used to produce the smaller table have their own costs; a storage reduction is not automatically an equal reduction in elapsed time.

Throwing away coordinates can erase an important difference. The points $(2, 1)$ and $(2, -1)$ have squared Euclidean distance 4. Orthogonally projecting onto the horizontal line sends both to $(2, 0)$, so their projected distance is zero. Projecting onto the vertical line preserves their full separation.

Explore a projection direction



Angle: 45°. Original squared distance: 4. Projected squared distance: 2.000.

This is ordinary orthogonal projection, without random-map rescaling. The PDF shows 45 degrees. At 0 degrees the squared distance is 0; at 90 degrees it is 4.

If the line makes angle θ with the horizontal axis, its unit direction is $(\cos\theta, \sin\theta)$. The pair difference is $(0, 2)$, so its coordinate along that line is $2\sin\theta$. The projected squared distance is therefore $4\sin^2\theta$. One direction can miss all of a difference; using several random directions distributes that risk.

2. One random matrix, shared by every observation

Store observations as rows of $X \in \mathbb{R}^{n \times d}$. Draw a matrix $R \in \mathbb{R}^{d \times k}$ once and compute:

$$Y = XR\,.$$

In words: The projected table Y equals X times R. The original table has n rows and d columns, the shared map has d rows and k columns, and the result has n rows and k columns.

Each output coordinate is a weighted sum of the original coordinates. For the Gaussian construction used here, entries of R are independent normal random variables with mean zero and variance $1 / k$. Equivalently, draw standard normal entries and divide by $\sqrt{k}$. These columns are not required to be orthogonal or unit length. "Random projection" is the standard name for this random linear map.

The same map is essential. For two row vectors x and y, linearity gives $xR - yR = (x - y)R$. Therefore, preserving a pairwise distance reduces to preserving the length of its difference vector. Independent maps for different observations would place their outputs in unrelated coordinate systems.

Why divide by the square root of the output dimension?

Fix a difference vector v before drawing R. A single projected coordinate is a sum of independent zero-mean terms. Its variance is $\|v\|^2 / k$. Summing the squared values of k coordinates gives the correct expected squared length:

$$\mathbb{E}\left\|vR\right\|^2 = \sum_{j=1}^{k} \frac{\|v\|^2}{k} = \left\|v\right\|^2.$$

In words: The expected squared length of the projected vector equals its original squared length. Each of the k projected coordinates contributes one kth of the original squared length in expectation.

Without the scale factor, the expected squared length would be k times too large. Correct expectation alone is not a guarantee: an estimate can have the right mean yet vary widely. Concentration bounds describe how increasing k controls that variation.

Construct and apply a Gaussian map

```
import numpy as np

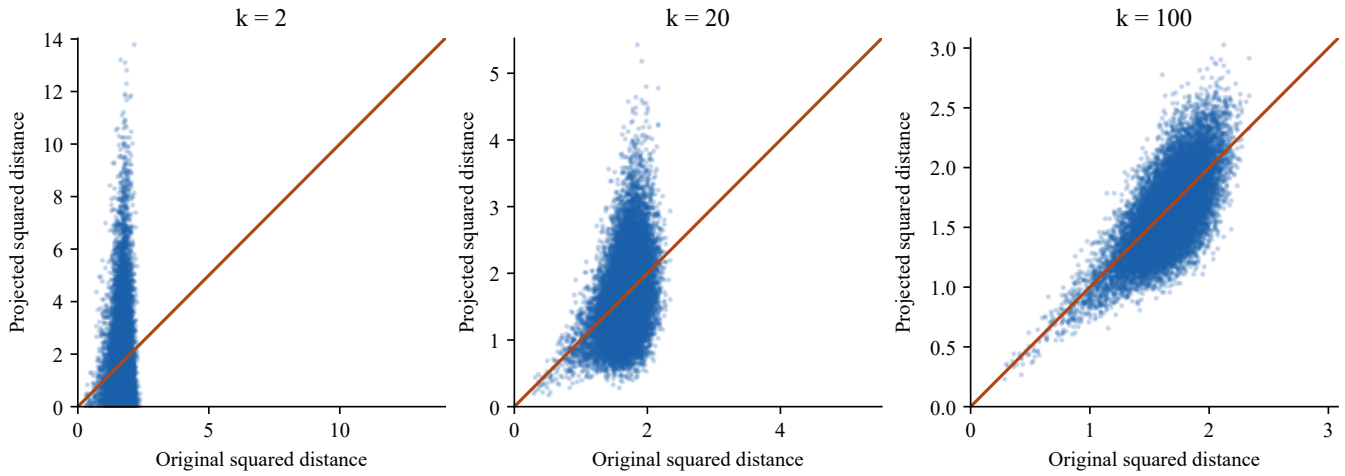
# X has one observation per row.
n, d = X.shape
k = 30
rng = np.random.default_rng(17)
R = rng.standard_normal((d, k)) / np.sqrt(k)
Y = X @ R
assert Y.shape == (n, k)
```

Save the map, or enough information to reproduce it exactly, when later points must use the same coordinates. A seed by itself is meaningful only together with the generator, construction, and coordinate order.

3. Measure the error of every pair

The course experiment uses 200 selected GloVe word vectors with 300 coordinates each, normalized to unit length before projection. It also provides synthetic data with different structure. The word example is a finite selected vocabulary, not a search over every English word. A cosine comparison of the normalized original vectors is equivalent to an ordering by Euclidean distance, because squared distance is $2(1 - \text{cosine})$.

There are $n(n-1)\,/\,2$ unordered pairs of distinct observations. For 200 points, that is **19,900 pairs**. Each dot in the following figure represents a pair, not a word. The horizontal coordinate is its original squared distance; the vertical coordinate is its projected squared distance. The diagonal is exact preservation.



Computed from the deck's word vectors. NumPy's default_rng seed 17 generates a 300 by 200 standard normal matrix; each panel uses its first k columns divided by square root of k. These maps are nested across k and need not match the slide deck's JavaScript generator. All points are shown; read the axis scales as well as the visual spread.

For a nonzero original distance, define the ratio q and absolute relative error e:

$$q = \frac{\|xR - yR\|^2}{\|x - y\|^2}, \qquad e = |\,q - 1\,|\,.$$

In words: The ratio is projected squared distance divided by original squared distance. Absolute relative error is the absolute value of that ratio minus one.

A ratio of 1 is perfect, 1.2 means the squared distance grew by 20%, and 0.8 means it shrank by 20%. Duplicate original points have zero difference and remain identical under a shared linear map; their ratio is undefined, so do not divide by zero when reporting relative errors.

Output dimension k	Median pair error	Worst pair error	Pairs within 0.5 tolerance
2	0.630	7.190	38.4%
20	0.214	1.967	89.9%
100	0.092	0.739	99.9%

The median describes a typical pair. The maximum decides whether *every* pair meets the tolerance. Even 99% passing would not satisfy an all-pairs requirement. These numbers describe the displayed fixed data and maps. They are measurements, not estimated universal failure probabilities.

Squared distances and ordinary distances

A tolerance $\varepsilon = 0.5$ on squared distances means the ratio of ordinary distances lies between $\sqrt{0.5}$ and $\sqrt{1.5}$, approximately 0.707 and 1.225. It does not mean the ordinary distance ratio lies between 0.5 and 1.5. State which quantity the tolerance controls.

Change a seed without changing the data

A new map can improve some pairs and worsen others. Increasing the output dimension improves reliability in the probabilistic analysis, but the observed error of every specific pair need not decrease at every step. A collection of independent trials helps distinguish a typical outcome from one lucky draw. The deck's 40-trial experiment reports empirical variation; 36 successes out of 40 is not a theorem that the next map succeeds with probability exactly 90%.

There is one useful deterministic monotonicity statement. If we keep the map fixed and enlarge a dataset by adding points, all previous pairs remain. Its worst pair error therefore cannot decrease. Its median error can move in either direction.

4. The Johnson-Lindenstrauss guarantee

Fix a finite set of n points independently of the random map. Choose an error tolerance $0 < \varepsilon < 1$ and failure probability $0 < \delta < 1$. Suitable random linear maps into $k = O(\varepsilon^{-2}\log(n / \delta))$ coordinates preserve all squared pairwise distances within factors $1 - \varepsilon$ and $1 + \varepsilon$ with probability at least $1 - \delta$. This randomized form of the Johnson-Lindenstrauss guarantee does not assume the data lies near a low-dimensional plane. See Dasgupta and Gupta for the concentration and union-bound approach.

The dependence on n is logarithmic, while requiring a smaller error can increase the required dimension substantially. The bound's constants matter in applications. A sufficient bound can exceed the original dimension; in that case it does not certify useful compression, even if a particular experiment performs well at a smaller dimension.

One pair, then every pair

For the Gaussian construction, a fixed nonzero vector's projected squared-length ratio has the distribution of the sum of k independent squared standard normals, divided by k. A standard two-sided concentration bound gives:

$$\Pr\left(\,|\,q - 1\,|\, > \varepsilon\right) \leq 2\exp\left[-k\left(\frac{\varepsilon^2}{4} - \frac{\varepsilon^3}{6}\right)\right].$$

In words: The chance that a fixed vector's squared-length ratio differs from one by more than epsilon is at most twice the exponential of minus k times epsilon squared over four minus epsilon cubed over six.

Let $M = n(n-1)/2$. The probability that at least one pair fails is at most the sum of the M pair-failure probabilities. This is the **union bound**. The pair events share a map and are not generally independent; independence is not required for this bound.

It is enough to make the fixed-pair bound at most δ / M. Solving that inequality gives the following sufficient dimension for $n \geq 2$:

$$k \geq \left\lceil \frac{\ln(2M/\delta)}{\varepsilon^2/4 - \varepsilon^3/6} \right\rceil.$$

In words: A sufficient output dimension is the ceiling of the natural logarithm of two times the number of pairs divided by delta, divided by epsilon squared over four minus epsilon cubed over six.

> **A conservative bound is not a prediction**
>
> For 200 points, squared-distance tolerance 0.5, and failure probability 0.05, the displayed sufficient bound gives $\mathbf{k = 327}$. The original vectors have only 300 coordinates, so this bound does not certify a smaller representation for those settings. The experimental table can still show useful errors at much smaller k. "This sufficient bound does not certify it" and "it cannot work" are different statements.

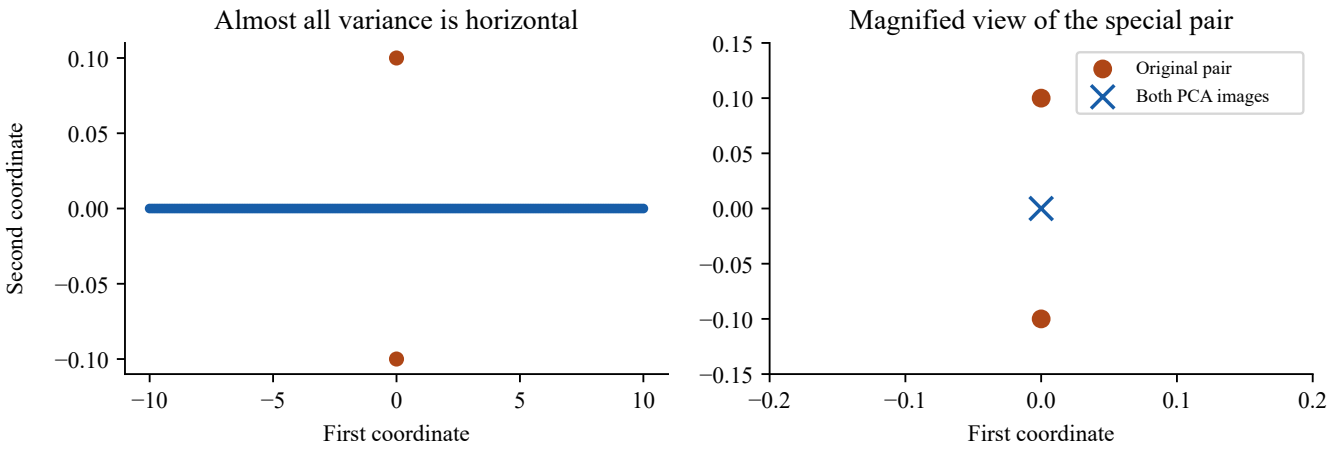
Why a nonzero nullspace does not contradict the guarantee

Any linear map from d coordinates to fewer than d coordinates has a nonzero kernel: some nonzero vector maps to zero. Thus no such map preserves the length of every vector in the entire original space. For example, $(x_1, x_2) \mapsto x_1 + x_2$ erases $(1, -1)$.

The guarantee fixes a finite dataset before drawing the map. Choosing an adversarial pair after inspecting the map changes the order of those choices. The theorem does not promise protection for every future point, an unlimited sequence of queries, or data adaptively constructed to exploit the sampled map.

5. Compare objectives, not just dimensions

Principal component analysis (PCA) centers the data and finds orthogonal directions capturing the largest total variance. Retaining the leading coordinates minimizes total squared reconstruction error among subspaces of that dimension. That global objective differs from controlling the largest relative error of any pair.



A constructed counterexample. The right panel magnifies the close pair; the two panels use different scales. PCA preserves almost all total variance with one horizontal coordinate, while losing the entire separation of the orange pair.

In this example, 198 points lie evenly along the horizontal axis from -10 to 10. Two additional points are $(0, 0.1)$ and $(0, -0.1)$. The mean is zero and the leading PCA direction is horizontal. It retains **99.99970%** of the total squared variation. Yet the special pair's squared distance falls from 0.04 to zero: a 100% relative loss for that pair.

PCA solved its optimization problem; that problem was not all-pairs relative-distance preservation. Conversely, a Gaussian map is not guaranteed to beat PCA on every dataset or measure. On an exact two-dimensional affine plane, centering and keeping its two PCA directions preserves all pairwise distances, up to numerical error. A random map at the same dimension generally distorts them.

Preserved distances can still change a nearest neighbor

Suppose two candidates are almost tied in distance from a query. A small upward error for one and downward error for the other can swap them. An all-pairs distance guarantee therefore does not imply an identical nearest-neighbor ranking.

A useful sufficient condition makes the gap explicit. If the original squared distances are $a < b$ and $(1+\varepsilon)a < (1-\varepsilon)b$, then allowed distortion cannot reverse their order. Without enough separation, rankings may change while both distances remain within tolerance. In the word experiment, neighbor overlap is an application-specific measurement among the selected 200 candidates.

6. Maintain the projection as events arrive

A stream can be represented by a count vector. Choose one coordinate for each event category; an occurrence adds 1 to that coordinate. Word counts are one example. The count vector already discards event order and timing, so no projection of it can recover that information.

Let the current row count vector be x and its **sketch** be $z = xR$. If an event changes coordinate i by Δ, then only one row of the map is needed to update the sketch:

$$z \leftarrow z + \Delta R_{i,:}.$$

In words: When coordinate i of the count vector changes by delta, add delta times row i of the shared map to the sketch.

This follows by distributing the matrix product over the updated count vector. It is an exact algebraic identity, apart from floating-point rounding. Positive events, deletions, and corrections use the same rule with different signs of Δ.

A hand-checkable stream

Use three coordinates in the order `[data, matrix, graph]` and four sketch coordinates. To keep arithmetic visible, choose the following illustrative sign matrix. Each entry is plus or minus one half. This particular displayed matrix is chosen for explanation; it is not presented as a randomly sampled map with an all-input guarantee.

```
R = [[ 0.5,  0.5, 0.5,  0.5],   # data
     [ 0.5, -0.5, 0.5, -0.5],   # matrix
     [-0.5,  0.5, 0.5, -0.5]]   # graph
```

Process `data, matrix, data, graph, matrix, data`, beginning with zero counts and a zero sketch. Each row below is calculated both by incremental updates and by multiplying the complete count vector by R; the results agree.

Step	Event	Count vector	Sketch
1	data	[1, 0, 0]	[0.5, 0.5, 0.5, 0.5]
2	matrix	[1, 1, 0]	[1.0, 0.0, 1.0, 0.0]
3	data	[2, 1, 0]	[1.5, 0.5, 1.5, 0.5]
4	graph	[2, 1, 1]	[1.0, 1.0, 2.0, 0.0]
5	matrix	[2, 2, 1]	[1.5, 0.5, 2.5, -0.5]
6	data	[3, 2, 1]	[2.0, 1.0, 3.0, 0.0]

The full counts are shown to check the arithmetic. The incremental update does not need them: it needs the event's coordinate, its change, and the corresponding map row. Independent random signs scaled by $1 / \sqrt{k}$ also provide a valid random-projection construction with suitable dimension; see Achlioptas, Database-friendly random projections.

Update a sketch without storing the full count vector

```
z = np.zeros(k)
for category_index, delta in events:
    z += delta * R[category_index, :]
```

Every worker must use the same category order. If coordinate 0 means “data” on one worker and “graph” on another, the same numeric matrix is not enough to make their sketches compatible.

7. Merge sketches and compare streams

Two workers with count vectors x and y can combine their sketches by addition if both use the same map and coordinate definitions:

$$(x + y)R = xR + yR\,.$$

In words: Projecting the sum of two count vectors equals the sum of their sketches, provided both sketches use the same map.

For example, splitting the displayed stream into two disjoint batches gives local counts `[2, 1, 0]` and `[1, 1, 1]`. Their sketches are `[1.5, 0.5, 1.5, 0.5]` and `[0.5, 0.5, 1.5, -0.5]`. The sum is `[2, 1, 3, 0]`, exactly the final sketch of the combined counts `[3, 2, 1]`.

If the workers choose different random maps, direct addition generally has no such interpretation. Reproducibility and common coordinate definitions are part of the computation, not merely housekeeping.

Subtraction works as well: $xR - yR = (x - y)R$. The squared length of that sketch difference can estimate the squared distance between the original count vectors. The merge and update identities are exact, while the distance estimate is probabilistic. One does not turn the other into an exact reconstruction.

A finite collection of stream prefixes, fixed independently of the map, can be treated as a finite set of vectors in a distance guarantee. Protecting more prefixes requires an appropriate failure budget. Successful behavior on a few observed prefixes establishes no guarantee for an unlimited or adaptively chosen future stream.

8. Account for the whole computation

Quantity or operation	Dense-map cost	What is being counted
Store the map	$O(dk)$	Map entries
Project one full vector	$O(dk)$	Arithmetic
Process one coordinate update	$O(k)$	Add a scaled map row
Store one running sketch	$O(k)$	Current sketch, in addition to map access
Store n projected observations	$O(nk)$	The entire projected dataset
Compare one pair of sketches	$O(k)$	Subtract and sum squared differences

The state of one running sketch does not grow with the number of events when d and k are fixed. Storing one sketch per observation is a different task and still grows with the number of observations. Map storage can sometimes be traded for reproducible generation, but generating map entries itself takes work and must preserve the intended distribution and consistency.

Decide which comparisons matter before choosing the dimension: name the dataset or stream summaries, the distance convention, the tolerance, and the failure probability. Then measure the errors and costs relevant to the task. A smaller representation earns its place by preserving a useful question.

Self-checks and worked answers

1. Why is a fresh map for every point wrong for this construction? Pair differences must be mapped by the same linear transformation. With different maps, subtracting the outputs does not equal the shared-map image of the input difference.

2. If a map works for every pair in one experiment, will the next map also work? Not necessarily. The guarantee concerns probability over draws, and the experiment observes one draw.

3. Does retaining 99.99% of PCA variance preserve every pair within 1%? No. The close-pair example loses an entire separation in a direction that contributes very little total variance.

4. Do pair-failure events have to be independent for the union bound? No. The probability of any failure is at most the sum of individual failure probabilities even when events are dependent.

5. What changes when an event count decreases by two? Subtract twice the appropriate row of R from the sketch. The same linear update formula applies.

6. Does an exact sketch merge imply exact distances? No. The merge yields exactly the sketch of the combined vector, up to arithmetic rounding. That sketch can still distort geometric quantities.

Further reading and data

Dasgupta and Gupta's elementary proof develops a concentration argument for Johnson-Lindenstrauss embeddings. Achlioptas's binary-coin constructions study simple random maps. Woodruff's Sketching as a Tool for Numerical Linear Algebra surveys broader uses of linear sketches. The selected word vectors come from the course copy of GloVe; synthetic examples and generator settings are identified where used.